YIELD DEVIATION INVESTIGATION ? API MANUFACTURING

Deviation Reference: DEV-2026-127
Title: Crystallization yield below specification for Ibuprofen API
Product: Ibuprofen API
Batch: IBU-26-0033
Manufacturing Stage: Crystallization and centrifugal filtration
Equipment or Area: Crystallizer CR-12 and centrifuge CF-07
Incident Date: 2026-10-08
Reported By: Tomasz Nowak (Production Supervisor)
Severity Suggested: Major
Detected By: yield calculation during batch record review
Quantity Affected: 1 batch (980 kg theoretical)
Immediate Containment: Batch IBU-26-0033 placed on QA hold; no release until the deviation is closed.

1. SUMMARY
The isolated yield of batch IBU-26-0033 after centrifugal filtration and drying was 87.2 % against the registered yield range of 93.0 % - 97.0 % (specification SPEC-IBU-011). The deviation was raised during batch record review before release.

2. PROCESS AND RECORD REVIEW
The batch manufacturing record shows the heptane anti-solvent was charged in 20 minutes on 2026-10-08 between 14:05 and 14:25, against the SOP-PR-022 charge time of 90 minutes. The crystal slurry was transferred to centrifuge CF-07 at 15:10. Mother liquor discharge volume recorded in the batch record was 1,240 L against a typical 980 L. No in-process control result was out of specification; the finished API met all quality attributes except yield.

3. DCS TREND REVIEW
The DCS trend for CF-07 on 2026-10-08 shows the centrifuge basket speed at 1,850 rpm during the whole filtration cycle, above the recipe setpoint of 1,200 rpm, and a wet cake thickness signal dropping sharply at 15:32. The filtration cycle ended 11 minutes earlier than the typical cycle time.

4. MAINTENANCE RECORDS
Maintenance work order WO-26-4531 reports that the filter cloth on CF-07 was found torn during post-batch inspection on 2026-10-08 at 17:50. The cloth was replaced the same day. The preventive maintenance record shows the CF-07 cloth inspection is scheduled monthly; the last recorded inspection was 2026-08-29. Spare cloth lot CLO-26-19 is recorded in the spare parts issue log.

5. PERSONNEL
Operator interview (2026-10-09, signed): "we increased the basket speed to keep the cycle time because the slurry was very fine; I did not check the recipe

limits". Training records show the operator was trained on SOP-PR-022 on 2026-03-18 and on centrifuge operation on 2026-03-20.

6. IMMEDIATE ACTIONS
- Batch IBU-26-0033 remains on QA hold for yield reconciliation.
- Basket speed alarm limits for CF-07 verified against the recipe on 2026-10-09.
- Filter cloth inspection after every batch until the deviation is closed.

Note: environmental monitoring data for Suite A on 2026-10-08 was within limits. No supplier or raw material issue was identified in this investigation.